COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	**CPE212 / Automating Server Management**	**SECTION:**	**CPE31S2**
DATE PERFORMED:	**Sept 24 2026**	**DATE SUBMITTED:**	**Sept 24 2026**
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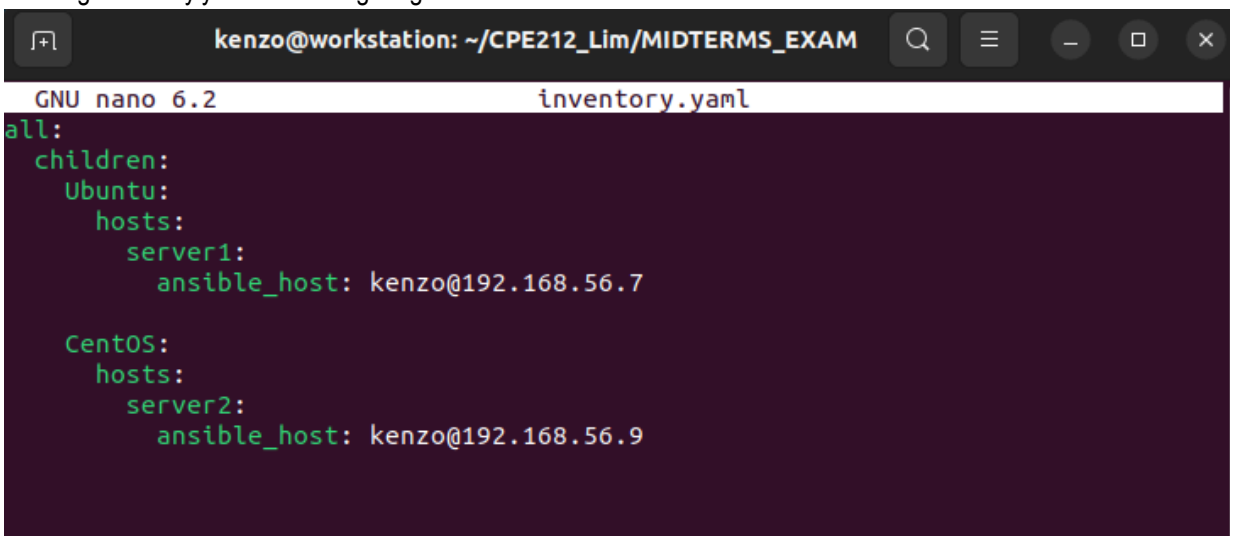
ACTIVITY NO. **Midterms Skills Exam**

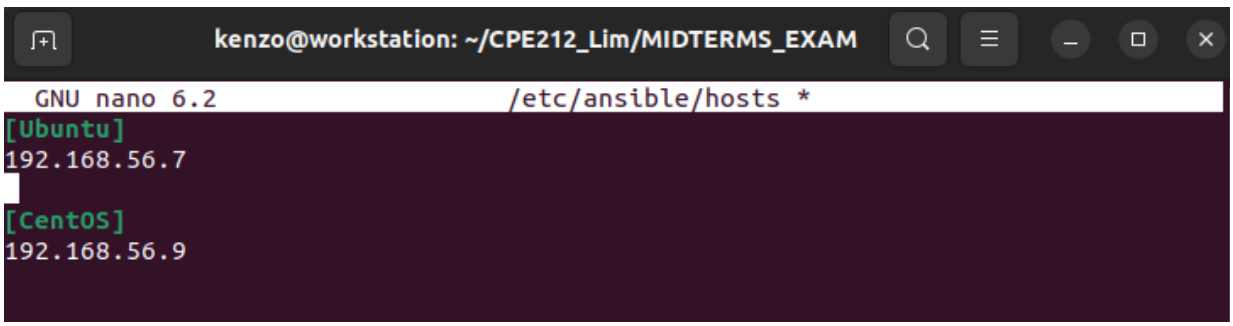
1. PROCEDURE OUTPUT

1. Create a playbook that installs Prometheus in both Ubuntu and CentOS. Apply the concept of creating roles.
2. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)
3. Show an output of the installed Prometheus for both Ubuntu and CentOS.
4. Make sure to create a new repository in GitHub for this activity.

Task 1: Prepare the Server

Creating Inventory.yaml and configuring /etc/ansible/hosts





Creating Roles for the playbook

```
kenzo@workstation:~/CPE212_Lim/MIDTERMS_EXAM$ ansible-galaxy init prometheus
- Role prometheus was created successfully
kenzo@workstation:~/CPE212_Lim/MIDTERMS_EXAM$ tree
.
├── inventory.yaml
└── prometheus
    ├── defaults
    │   └── main.yml
    ├── handlers
    │   └── main.yml
    ├── meta
    │   └── main.yml
    ├── README.md
    ├── tasks
    │   └── main.yml
    ├── tests
    │   ├── inventory
    │   └── test.yml
    └── vars
        └── main.yml
```

Update the package repository.

kenzo@workstation: ~/CPE212_Lim/MIDTERMS_EXAM

GNU nano 6.2 install_prometheus.yaml

```yaml
---
- hosts: all
  become: true
  pre_tasks:
   # Task 1 : Prepare the server
   - name: Update package repository for Ubuntu
     ansible.builtin.apt:
       update_cache: yes
     when: ansible_distribution == "Ubuntu"

   - name: Update package repository for CentOS
     ansible.builtin.dnf:
       update_cache: yes
     when: ansible_distribution == "CentOS"
```

Upgrade installed packages.

```
# Upgrade Installed Packages
- name: Upgrade Installed Packages for Ubuntu
  ansible.builtin.apt:
    upgrade: dist
  when: ansible_distribution == "Ubuntu"

- name: Upgrade Installed Packages for CentOS
  ansible.builtin.dnf:
    update_only: true
  when: ansible_distribution == "CentOS"
```

Install the utilities required by the installation.

```
# Install utilities required by the installation
- name: Install utilities required by the Installation (Ubuntu)
  ansible.builtin.apt:
    name:
      - tar
      - wget
      - curl
  when: ansible_distribution == "Ubuntu"

- name: Install utilities required by the Installation (CentOS)
  ansible.builtin.dnf:
    name:
      - tar
      - wget
      - curl
  when: ansible_distribution == "CentOS"
```

In this part, I began setting up the Servers by creating an inventory file and initializing ansible roles, as well as adding the servers 1 and 2, after that I then updated and upgraded package repositories of the two managed nodes.

Task 2: Create the Prometheus User

1. Create a dedicated prometheus system user.
2. Prevent the user from having a normal login shell.
3. Verify that the user was successfully created.

```
kenzo@workstation: ~/CPE212_Lim/MIDTERMS_EXAM/prom...
 GNU nano 6.2                          main.yml
---
# tasks file for prometheus
  # Task 2: Create the Prometheus User
  - name: Create a dedicated prometheus system user and prevent from having
    ansible.builtin.user:
      name: prometheus
      system: true
      create_home: false
      shell: /bin/false
      state: present

  - name: Verify that the user was successfully created
    ansible.builtin.command: id prometheus
```

Here in this part, I was able to create the prometheus user and stated shell: /bin/false to prevent the user from having a normal login shell

Task 3: Create Prometheus Directories

1. Create /etc/prometheus.
2. Create /var/lib/prometheus.
3. Assign the appropriate owner and group.
4. Verify the directory permissions and ownership.

```
  # Task 3: Create Prometheus Directories
  - name: Create Prometheus Directories and Assign owner and group
    ansible.builtin.file:
      path: /etc/prometheus
      state: directory
      owner: prometheus
      group: prometheus
      mode: "0755"

  - name: Create libraries
    ansible.builtin.file:
      path:  /var/lib/prometheus
      state: directory
      owner: prometheus
      group: prometheus
      mode: "0755"
```

```
[kenzo@server3 ~]$ ls -ld /etc/prometheus /var/lib/prometheus
drwxr-xr-x. 2 prometheus prometheus 6 Sep 24 10:03 /etc/prometheus
drwxr-xr-x. 2 prometheus prometheus 6 Sep 24 10:03 /var/lib/prometheus
```

Here, I created prometheus directories and assigned owner and group permissions and verified them in the managed nodes

Task 4: Download Prometheus

Download the Prometheus archive.

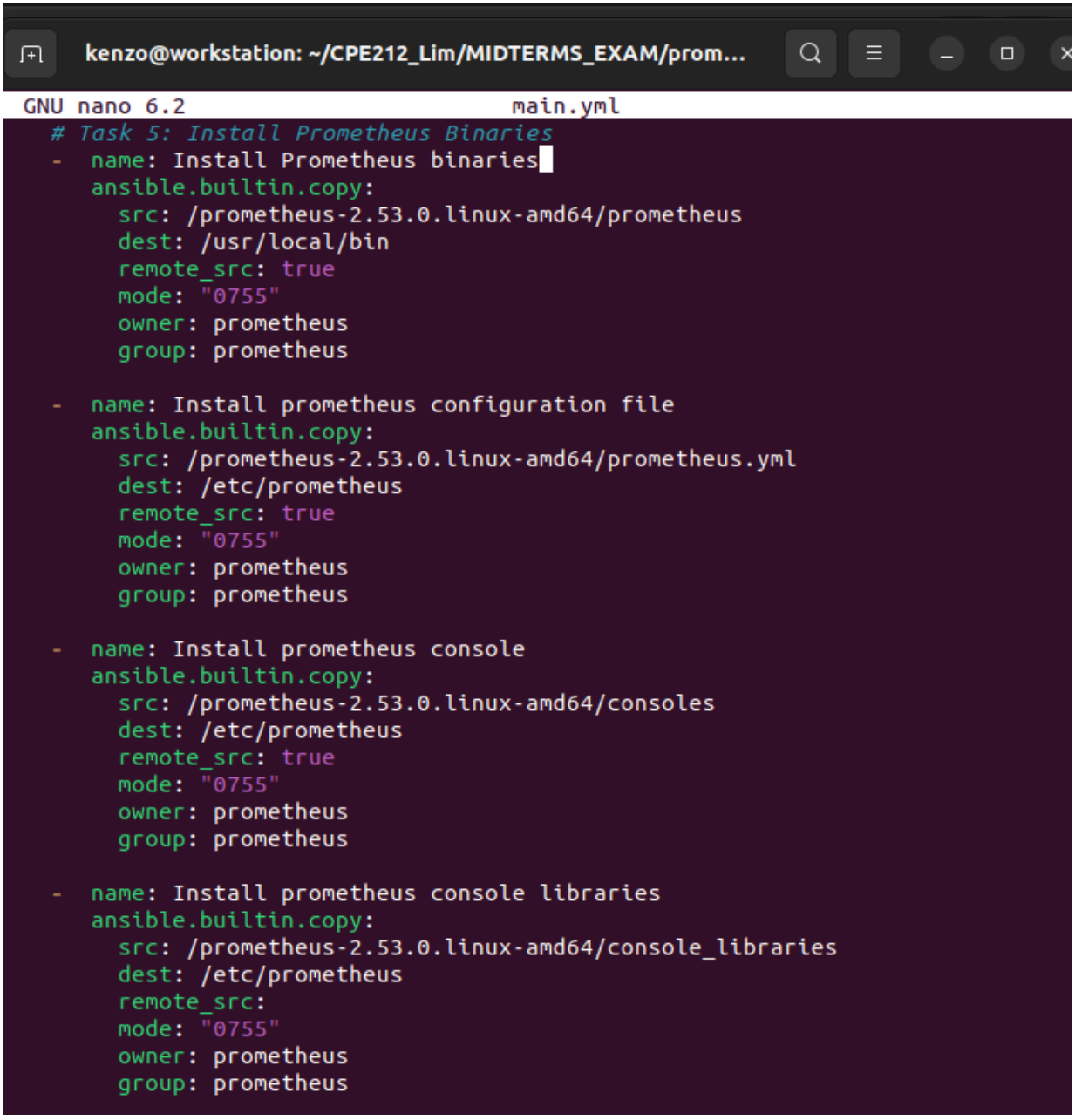
Extract the Prometheus archive.

```
# Task 4: Download Prometheus
-  name: Download Prometheus archive
   ansible.builtin.get_url:
     url: https://github.com/prometheus/prometheus/releases/download/v2.53.>
     dest: /
     mode: "0755"

-  name: Extract Prometheus archive
   ansible.builtin.unarchive:
     src: /prometheus-2.53.0.linux-amd64.tar.gz
     dest: /
     remote_src: true
     mode: "0755"
     owner: prometheus
     group: prometheus
```

In this part I downloaded and extracted prometheus archive inside the / directory.

Task 5: Install the Prometheus Binaries



In this part, I copied some of the prometheus libraries to /etc/prometheus, these include the prometheus.yml and console/console_libraries.

Task 6: Install the Prometheus Configuration Files

Task 7: Validate the Configuration

Task 8: Create the systemd Service

Task 9: Start and Enable Prometheus

Task 10: Verify Prometheus

Task 11: Access Prometheus Through a Browser

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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